

Shashank Gupta

E-mail: 27392shashankgupta@gmail.com
Github: <https://github.com/shashankg7>

Google Scholar Profile: <https://bit.ly/2E6bw7W>
Web-Page: <http://shashank-gupta.com/>

RESEARCH INTERESTS Machine learning, Information retrieval (IR), Off-policy learning for search, recommender systems, and post-training foundational models.

PROFESSIONAL EXPERIENCE **Microsoft AI, Bangalore, India**
Senior Applied Scientist, Advertisements Team, April 2026 - Current
Machine learning for ads.

Meta AI, New York, USA
Research Scientist Intern, Modern Recommender Systems Team, Aug 2024 - Dec 2024
Mentors: Satya Narayan Shukla, Chaitanya Ahuja, Tsung-Yu Lin, and Sreya Dutta Roy.
Worked on a novel reinforcement learning method for post-training text-to-image diffusion models. Resulted in a paper accepted at [Transactions on Machine Learning Research \(TMLR\)](#), 2026.

Meta AI, London, UK
Research Scientist Intern, Modern Recommender Systems Team, Aug 2023 - Dec 2023
Mentors: Eric Tan, and Yiming Liao.
Worked on off-policy learning for the two-stage recommender system, and on mixture of experts network for the recommender system. Resulted in a paper accepted at [ICTIR 2025 \(co-located with SIGIR 2025\)](#).

Flipkart, Bangalore, India
Data Scientist, Search Ranking Team, July 2018 - Jan 2021
Mentor: Mohit Kumar, and Narendra Varma.
Worked on learning-to-rank methods for personalized fashion search, resulted in $X\%$ improvement in the conversion rate. Worked with terabyte scale log data using PySpark and Tensorflow. Also worked on the problem of session length prediction for user's incoming search session to help disambiguate their [explore v/s purchase intent \(SIGIR'20\)](#), and [user return time prediction using neural point process model \(UMAP'22 Workshop\)](#).

Conduent Labs (erstwhile Xerox Research (XRCI)), Bangalore, India
Research Internship, Jan 2018 - May 2018
Mentors: Manjira Sinha, and Sandya Mannarswamy.
Worked on the problem of fake news detection. Proposed a novel Coupled Matrix-Tensor Factorization (CMTF) based solution for the problem ([ASONAM'18](#)).

Tata Research, Design and Development Center, Pune, India
Research Internship, May 2017 - July 2017
Mentors: Girish Palshikar, Sachin Pawar, and Nitin Ramrakhiani.
Worked on semi-supervised and multi-task learning based methods for Adverse Drug Reaction (ADR) mention extraction from social media posts ([ECIR'18](#), [CIKM'17](#) and [NeurIPS'17 workshop](#)).

EDUCATION **University of Amsterdam (UvA)**
PhD, ML and IR, April 2021 - October 2025
Advisors: Dr. Maarten de Rijke, UvA
Dr. Harrie Oosterhuis, Radboud University
Research Group: **The Information Retrieval Lab (IRLab)**.

International Institute of Information Technology (IIIT), Hyderabad
Master of Science By Research, Computer Science and Engineering, June 2015 - Dec'17
Advisors: Dr. Vasudeva Varma, Dean R&D, IIIT-Hyderabad
Dr. Manish Gupta, Principal Applied Scientist, Microsoft, Hyderabad
Research Group: Information Retrieval and Extraction Laboratory (IREL).

Birla Institute of Technology, Mesra, India
Bachelor Of Technology, Computer Science and Engineering, June 2010 - June 2014.

TUTORIALS

[Unbiased Learning to Rank: On Recent Advances and Practical Applications](#)

[Shashank Gupta](#), Philipp Hager, Jin Huang, Ali Vardasbi, and Harrie Oosterhuis.
WSDM 2024.

[Recent Advances in the Foundations and Applications of Unbiased Learning to Rank](#)

[Shashank Gupta](#), Philipp Hager, Jin Huang, Ali Vardasbi, and Harrie Oosterhuis.
SIGIR 2023.

[Recent Advancements in Unbiased Learning to Rank](#)

[Shashank Gupta](#), Philipp Hager, and Harrie Oosterhuis.
FIRE 2023.

PUBLICATIONS

[Variance-Optimal Action Logging and Learned Control Variates for Sample-Efficient Policy A/B Tests](#)

[Shashank Gupta](#), Subhadeep Maji.
Under review.

[A Simple and Effective Reinforcement Learning Method for Text-to-image Diffusion Model](#)

[Shashank Gupta](#), Chaitanya Ahuja, Tsung-Yu Lin, Sreya Dutta Roy, Harrie Oosterhuis, Maarten de Rijke, and Satya Narayan Shukla.
Transactions on Machine Learning Research (TMLR) 2026.

[Additive Control Variates Dominate Self-Normalisation in Off-Policy Evaluation](#)

Olivier Jeunen, [Shashank Gupta](#).
SIGIR 2026.

[Safe, Efficient, and Robust Reinforcement Learning for Ranking and Diffusion Models](#)

[Shashank Gupta](#).
PhD thesis. Thesis abstract also published at the [SIGIR Forum 2025](#).

[Towards Two-Stage Counterfactual Learning to Rank](#)

[Shashank Gupta](#), Yiming Liao, and Maarten de Rijke.
ICTIR 2025 (Co-located with SIGIR).

[Practical and Robust Safety Guarantees for Advanced Counterfactual Learning to Rank](#)

[Shashank Gupta](#), Harrie Oosterhuis, and Maarten de Rijke
CIKM 2024 (**Full paper**).

[Optimal Baseline Corrections for Off-Policy Contextual Bandits](#)

[Shashank Gupta*](#), Olivier Jeunen*, Harrie Oosterhuis, and Maarten de Rijke (***Equal Contribution**)
RecSys 2024 (**Full paper**).

Proximal Ranking Policy Optimization for Practical Safety in Counterfactual Learning to Rank

Shashank Gupta, Harrie Oosterhuis, and Maarten de Rijke
CONSEQUENCES @RecSys 2024 (**Oral presentation**).

A Simpler Alternative to Variational Regularized Counterfactual Risk Minimization

Hua Chang Bakker, Shashank Gupta, and Harrie Oosterhuis
CONSEQUENCES @RecSys 2024 (**Poster presentation**).

Safe Deployment for Counterfactual Learning to Rank with Exposure-Based Risk Minimization

Shashank Gupta, Harrie Oosterhuis, and Maarten de Rijke
SIGIR 2023 (**Full paper**), CONSEQUENCES at RecSys 2023.

A Deep Generative Recommendation Method for Unbiased Learning From Implicit Feedback

Shashank Gupta, Harrie Oosterhuis, and Maarten de Rijke
ICTIR 2023, CONSEQUENCES+REVEAL at RecSys 2022.

A First Look at Selection Bias in Preference Elicitation for Recommendation

Shashank Gupta, Harrie Oosterhuis, and Maarten de Rijke
CONSEQUENCES at RecSys 2023.

Neural Bag-of-Words Point Process Model for User Return Time Prediction in E-commerce

Shashank Gupta, Manish Bansal
CRUM Workshop at UMAP 2023.

The University of Amsterdam at the TREC 2021 Fair Ranking Track

Ali Vardasbi, Gabriel Bénédict, Shashank Gupta, Maria Heuss, Pooya Khandel, Ming Li, Fatemeh Sarvi
TREC Fair Ranking Trec 21.

Predicting Session Length for Product Search on E-commerce Platform

Shashank Gupta, Subhadeep Maji
SIGIR 2020, Short.

On Application of Bayesian Parametric and Non-parametric Models for User Cohorting in Product Search

Shashank Gupta
ECNLP@ACL20, Position paper.

RARE: A Recurrent Attentive Recommendation Engine for News Aggregators

Vaibhav Kumar, Dhruv Khattar, Shashank Gupta, Manish Gupta, and Vasudeva Varma
CIKM 2018, Workshop on News Recommendation and Analytics (INRA 2018).

CIMTDetect: A Community Infused Matrix-Tensor Coupled Factorization Based Method for Fake News Detection

Shashank Gupta, Raghuveer Thirukovalluru, Manjira Sinha, and Sandya Mannar-swamy.
ASONAM 2018, Short.

A Comparative Study of Embeddings Methods for Hate Speech Detection from Tweets

Shashank Gupta, Zeerak Waseem

EMNLP 2018 Workshop on Noisy User Generated Text (Accepted as a poster)

Multi-Task Learning for Extraction of Adverse Drug Reaction Mentions from Tweets

Shashank Gupta, Manish Gupta, Vasudeva Varma, Sachin Pawar, Nitin Ramrakhiyani, Girish Keshav Palshikar

ECIR 2018 (Full) and NIPS 2017 ML4H Workshop.

A Co-training Based Method for Extraction of Adverse Drug Reaction Mentions from Tweets

Shashank Gupta, Manish Gupta, Vasudeva Varma, Sachin Pawar, Nitin Ramrakhiyani, Girish Keshav Palshikar

ECIR 2018 (Short) and NIPS 2017 ML4H Workshop.

TCS Research at TAC 2017: Joint Extraction of Entities and Relations from Drug Labels using an Ensemble of Neural Networks

Sachin Pawar, Nitin Ramrakhiyani, Girish Keshav Palshikar, Shashank Gupta, Vasudeva Varma

TAC 2017 ADR Track.

Semi-Supervised Recurrent Neural Network for Adverse Drug Reaction Mention Extraction

Shashank Gupta, Sachin Pawar, Nitin Ramrakhiyani, Girish Keshav Palshikar, and Vasudeva Varma

BMC Bioinformatics Special Issue, and CIKM 2017, Workshop.

Enhancing Categorization of Computer Science Research Papers using Knowledge Bases

Shashank Gupta, Priya Radhakrishnan, Manish Gupta, Vasudeva Varma

SIGIR 2017, Works. on Knowledge Graphs & Semantics for Text Retrieval & Analysis.

Deep Neural Architectures for News Recommendation,

Vaibhav Kumar, Dhruv Khattar, Shashank Gupta, Manish Gupta, Vasudeva Varma
Conference and Labs of the Evaluation Forum (CLEF) 2017 (50 citations).

Simultaneous Inference of User Representations and Trust,

Shashank Gupta, Pulkit Parikh, Manish Gupta, Vasudeva Varma

ASONAM 17, Short.

Deep Learning for Hate Speech Detection in Tweets

Pinkesh Badjatiya*, Shashank Gupta*, Manish Gupta, Vasudeva Varma (***Equal Contribution**)(**Best Poster Award**)

WWW 2017, Poster track.

Scientific Article Recommendation by using Distributed Representations of Text and Graph

Shashank Gupta, Vasudeva Varma

WWW 2017, Workshop track.

User Profiling based Deep Neural Network for Temporal News Recommendation

Vaibhav Kumar, Dhruv Khattar, Shashank Gupta, Manish Gupta, and Vasudeva Varma

ICDM 2017, Workshop Track.

Word Semantics based 3D Convolutional Neural Networks for News Recommendation

Vaibhav Kumar, Dhruv Khattar, Shashank Gupta and Vasudeva Varma

ICDM 2017, Workshop track.

AWARD AND ACHIEVEMENTS	Received the Best Poster Paper Award at International World Wide Web Conference (WWW) 2017. Work on hate speech detection got covered in The Hindu, Indian Express and few other leading news outlets https://tinyurl.com/3p82typm. SIGIR student travel grant to attend SIGIR and ICTIR 2023. SIGCHI student travel grant to attend UMAP 2023 (declined). Travel Grant to attend NIPS 2017 (declined). TCS Research travel grant for attending CIKM 2017. Selected to attend IISC's Winter School on Machine Learning 2015.
SKILLS & TOOLS	PyTorch, Keras, Scikit-Learn, Tensorflow, Theano Lucene, ElasticSearch Python, MATLAB, C, C++, Java
PROFESSIONAL ACTIVITIES	Reviewer: SIGIR(21-24), ECIR(19-23), KDD'24, NeurIPS(20-23), ICML(20-24), ICLR(21-24), ACL(21), AAAI'21, EMNLP(20-21), CIKM'21, ACL'21, IJCNLP'20, ML4H@NIPS. Teaching Assistant: RecSys, UvA (2023/24), Advanced Information Retrieval, UvA (2022), Information Retrieval and Extraction, IIIT-H (2018), Machine Learning, IIIT-Hyderabad (2017), Machine Learning, BITS-Pilani (2015).
RELEVANT COURSEWORK	Machine Learning Information Retrieval and Extraction Topics in Information Retrieval Topics in Natural Language Processing Digital Image Processing
STUDENT SUPERVISION	Hua Chang Bakker (BSc AI 2024): Variational Counterfactual Risk Minimization for Contextual Bandits. Richter Van Emmerik (BSc AI 2023): Exploring large language models for recommender system. Beer Meester (BSc AI 2023): Offline reinforcement learning for learning to rank. Srijan Kaur (Masters UvA 2022): Knowledge distillation for efficient user modeling.
TALKS	<i>Reinforcement Learning from User Interactions for search & recommendations</i>, Invited talk at Nykaa, Bangalore. <i>Reinforcement Learning from User Interactions for search & recommendations</i>, Invited talk at Ebay, USA. <i>Simple and Effective Reinforcement Learning for Recommender Systems and Diffusion Models</i>, Presented at LossFunk, Bangalore. <i>Safe Deployment for Counterfactual Learning to Rank</i>, Presented at Expedia, London (remotely). <i>Optimal Baseline Corrections for Off-policy Contextual Bandits</i>, Oral Presentation at RecSys 2024. <i>Proximal Ranking Policy Optimization for Safe Counterfactual Learning to Rank</i>, Presentation at CONSEQUENCES, RecSys 2024. <i>Unbiased Learning to Rank: On Recent Advances and Practical Applications</i>, Tutorial presentation at WSDM 2024. <i>Off-policy Evaluation and Learning for Contextual Bandits</i>, Lecture presentation at the RecSys course, UvA. <i>Safe Deployment for Counterfactual Learning to Rank</i>, Presented at ShareChat ML Seminar (remotely). <i>Recent Advancements in Counterfactual Learning to Rank</i>, Tutorial presentation at SIGIR 2023. <i>Recent Advancements in Counterfactual Learning to Rank</i>, Invited tutorial presentation at FIRE 2023. <i>Safe Deployment for Counterfactual Learning to Rank</i>, Presented at MetaAI, New York (remotely).

Safe Deployment for Counterfactual Learning to Rank, Oral presentation at SIGIR 2023.

VAE-IPS: A Deep Generative Recommendation Method for Unbiased Learning From Implicit Feedback, Presented at CONSEQUENCES+REVEAL workshop@RecSys'22.

Session Length Prediction for Product Search on E-commerce Platform, Presented at SIGIR, 2020 <https://dl.acm.org/doi/abs/10.1145/3397271.3401219>

Scientific Article Recommendation using Deep Embeddings, Presented at World Wide Web Conference, Perth, 2017 <http://bit.ly/2nfmhi3>

Trust Prediction in Social Network using Deep Neural Networks, Presented at ASONAM conference, Sydney, 2017 <http://bit.ly/2zzya7CF>

Semi-supervised Recurrent Neural Network for Adverse Drug Reaction Mention Extraction from Twitter, Presented at CIKM Conference, Singapore, 2017

Pharmacovigilance from Social Media using Multi-task and Semi-Supervised Learning, Presented at ECIR Conference, Grenoble, 2018

Machine Learning Methods for Mining Adverse Drug Reactions from Social Media, Presented at DUKE-NUS Medical School, Singapore, 2017

Deep Learning Methods for Recommendation Systems, Presented at Thiagarajar College of Engineering, Madurai

Information Retrieval from Social Media, Presented at Alumni Research Talk (ART), BITS Pilani, Pilani.